MINOR PROJECT REPORT

Student Performance Analysis and Prediction System

Python | Exploratory Data Analysis | Linear Regression

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AI/ML with Python Training Program

Dataset: UCI Student Performance (Portuguese course) | DOI: 10.24432/C5TG7T | CC BY 4.0

Abstract

This project analyzes academic records from the Portuguese-course subset of the UCI Student Performance dataset and builds a Linear Regression model to estimate final-period grade (G3). Inputs are weekly study-time category, number of absences, first-period grade (G1) and second-period grade (G2). It includes data inspection, cleaning checks, descriptive statistics, feature engineering, the requested EDA visualizations, training/testing, cross-validation, basic parameter search and regression evaluation.

1. Introduction and problem statement

Academic performance is associated with learning habits, attendance patterns and earlier academic results. This project uses historical student data to inspect those patterns and estimate a final examination mark. The system is an analysis aid, not a definitive assessment of an individual student.

2. Objectives

• Inspect and clean the student performance data.

• Calculate mean, median and standard deviation.

• Analyze studytime, absences, G1, G2 and G3.

• Generate bar, histogram, scatter, box and correlation heatmap visualizations.

• Train Linear Regression and evaluate with MAE, RMSE, R-squared and cross-validation.

• Interpret findings, limitations and possible improvements.

3. Dataset and source

The data is from the UCI Machine Learning Repository's Student Performance dataset, Portuguese-course file (649 records). Original authors: Paulo Cortez and Alice Silva (2008). DOI: 10.24432/C5TG7T. License: CC BY 4.0. The working CSV is a five-column extract: studytime, absences, G1, G2 and G3. See data/source_info.md for attribution.

Variable meanings: studytime is a weekly time band (1: <2 hours; 2: 2-5 hours; 3: 5-10 hours; 4: >10 hours); absences is an absence count (not attendance percentage); G1 and G2 are earlier-period grades; G3 is final-period grade on a 0-20 scale.

4. Data preprocessing and feature engineering

The selected columns contain 649 records, no missing values, and no missing target values. The complete 33-column original source was checked and contained zero exact duplicate rows. The five-column extract has 109 repeated feature combinations, retained because different students can share the same selected values. Study-time and absence groups were derived for EDA.

5. Exploratory data analysis

The mean G3 is 11.906, the median is 12.000 and standard deviation is 3.231. Correlations with G3 are G2: 0.919, G1: 0.826, studytime: 0.250, and absences: -0.091. Charts are provided in outputs/visualizations:

• bar_mean_grade_by_studytime.svg

• histogram_final_grade.svg

• scatter_studytime_vs_final_grade.svg

• scatter_absences_vs_final_grade.svg

• scatter_g2_vs_final_grade.svg

• boxplot_grade_by_studytime.svg

• correlation_heatmap.svg

6. Machine learning methodology

Target: G3. Predictors: studytime, absences, G1 and G2. An 80/20 split with random_state=42 produced 519 training rows and 130 held-out test rows. The model uses a Scikit-learn Pipeline with median imputation, standard scaling and Linear Regression. GridSearchCV tested fit_intercept=True and False using shuffled 5-fold cross-validation on the training partition only. Transformations are fitted within the pipeline to avoid test-data leakage.

7. Evaluation results

Metric	Result
Test MAE	0.748
Test RMSE	1.171
Test R-squared	0.859
5-fold CV MAE mean	0.816
5-fold CV MAE standard deviation	0.107
5-fold CV R-squared mean	0.841

The held-out predictions are typically within about 0.75 grade points on this split.

Sample actual versus predicted marks The following rows are examples from the held-out test partition. The complete 130-row prediction file is generated at outputs/evaluation_results/test_predictions.csv when the script is run.

Test row	Actual G3	Predicted G3	Absolute error
2	12	13.36	1.36
6	13	12.52	0.48
10	14	14.52	0.52
24	10	11.25	1.25
30	11	11.08	0.08
31	15	15.58	0.58
39	12	13.66	1.66
41	11	11.15	0.15
44	11	11.42	0.42
49	12	12.57	0.57

The complete project implementation writes individual actual and predicted G3 values to outputs/evaluation_results/test_predictions.csv after running src/model.py.

8. Findings and interpretation

G2 has the strongest linear relationship with G3, followed by G1. The study-time category has a smaller positive association, and absences have a weak negative association. These are observed relationships, not evidence of cause and effect. G1 and G2 are known only after earlier assessments and therefore make this a later-period prediction model.

9. Limitations and future improvements

• The dataset covers two Portuguese schools and one course; findings may not generalize to other education systems.

• Study time is recorded in broad weekly categories; absences do not give an attendance percentage.

• G1 and G2 are strong features but make the model unsuitable before those assessments.

• Correlations do not establish causation.

• Future work could add more schools and subjects, attendance rates where recorded, additional models, repeated splits and prediction uncertainty.

10. Conclusion

The project implements a complete regression workflow for the UCI Portuguese-course student data. It performs data checks, descriptive analysis, required visualizations, model training, cross-validation, tuning and evaluation. The held-out MAE is 0.748, RMSE is 1.171 and R-squared is 0.859 for this split. Outputs should support investigation and not act as the sole basis for educational decisions.

References

Cortez, P., & Silva, A. (2008). Using data mining to predict secondary school student performance. Proceedings of the 5th Future Business Technology Conference (FUBUTEC 2008), 5-12. UCI Student Performance dataset. https://doi.org/10.24432/C5TG7T. License: CC BY 4.0.